Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	31 January 2026
Team ID	
Project Name	Health360: Executive Analytics & Population Health Dashboard
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority (High/Medium/ Low)
1	Data Access & Integration	The solution must ingest and connect to the flat file database smart_healthcare_dataset.csv containing 5,000 synthetic patient records across clinical vitals, lifestyle habits, and demographics.	High
2	Data Processing	The solution must cleanse data, reclassify age into 6 chronological brackets (Children, Young Adult, Adult, Middle Age, Senior, Elder), calculate composite risk scores, and generate optimized .hyper data extracts.	High
3	Business Analysis	The solution must calculate executive KPIs (Total Patients = 5,000, Baseline Risk Score = 1.57) and measure co-occurrences across chronic diseases (Diabetes, Heart Disease, Stroke) and lifestyle behaviours.	High
4	Business Insights & Analytical Outputs	The solution must render 9 dedicated analytical worksheets, a single-screen Master Dashboard, and a 9-point executive Tableau Story for guided decision-making.	High
5	User Interaction & Analysis	The solution must enable global cross-filtering across all 9 dashboard worksheets, allowing executive personas (Dr. Carter, Emily Rodriguez, Dr. Priya Nair) to filter by age, BMI, and lifestyle habits with a single click.	High
6	Data Update & Reporting	The solution must publish to Tableau Public Server to support browser-based executive access, interactive web viewing, and export capabilities for executive presentations.	Medium

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	Intuitive grid layout following standard UI visual hierarchy, clear color coding for risk levels, tooltip hover cards, and self-explanatory interactive legends.
NFR-2	**Security**	Data governance compliance using fully anonymized synthetic health records without Personally Identifiable Information (PII).
NFR-3	**Reliability**	Consistent dataset rendering ensuring 100% data integrity and cross-filter alignment across all 9 connected worksheets without calculation mismatches.

NFR-4	**Performance**	Rapid initial workbook load time under 3 seconds and interactive cross-filter response latency under 1 second using Tableau Data Extract (.hyper) optimization.
NFR-5	**Availability**	Cloud-hosted on Tableau Public Cloud Engine, ensuring 24/7 web accessibility across desktop and tablet browsers without local software installation requirements.
NFR-6	**Scalability**	Data schema and extracted calculations engineered to seamlessly scale from 5,000 to 100,000+ patient records without requiring structural visual redesigns.